# Shiftless & Carryless Collatz Model:

## *Extraction of the Sierpiński Gasket and Emergent Chaos via Redundant Binary Representation*

*Hiroshi Harada — April 4, 2026*


## Abstract

The difficulty of analyzing the Collatz conjecture (the "$3n + 1$ problem") can be traced to two fundamental sources of information loss:

1. Nonlinear information destruction caused by right shifts (division by 2) in even steps
2. Carry propagation generated during the odd-step operation $3n + 1$

These two mechanisms obscure the underlying geometric structure of the trajectory.
In this study, we introduce three complementary abstractions:

- a **Shiftless model**, which eliminates right shifts and preserves all bit information
- a **Redundant Binary (RB)** representation, which retains carries instead of resolving them immediately
- a **Carryless model**, which removes carries entirely by projecting coefficients modulo 2

By mapping binary integers to polynomials, the multiplication by 3 becomes algebraically equivalent to multiplication by $(x + 1)$. Projecting this polynomial representation onto the finite field GF(2) yields exactly the same pattern as the one-dimensional cellular automaton Rule 90, known for generating the Sierpiński gasket.
We show that the complexity of the standard Collatz trajectory emerges from the interaction between:

- fractal order generated by $(x + 1)$,
- LSB noise injected by the position of the least significant bit, and
- carry avalanches that globally distort the structure.

This framework provides a new algebraic and visual explanation for the origin of chaos in the Collatz dynamics.


## 1. Introduction

Despite its simple definition, the Collatz conjecture exhibits highly complex trajectory behavior. This complexity arises primarily from:

1. Irreversible information loss due to right shifts (division by 2)
2. Carry propagation during the odd-step operation $3n + 1$

To isolate these two destructive mechanisms, we introduce two layers of abstraction:

- Shiftless transformation: remove right shifts
- Carryless transformation: remove carries

This decomposition allows us to reinterpret the Collatz trajectory as the interaction of three components: order × noise × carry.

## 2. Shiftless Model

The Shiftless model eliminates right shifts and preserves the full weight of the least significant bit (LSB):

$$n_{k+1} = 3n_k + LSB(n_k)$$

where $LSB(n) = 2^{\nu_2(n)}$ is the weight of the lowest 1-bit in the binary representation of $n$.

Although Shiftless is algebraically equivalent to the odd subsequence of the standard Collatz map, it preserves all bit information and reveals the continuous expansion of the trajectory (**Figure 1**).

## 3. Redundant Binary (RB) Representation and Carryless Model

The core of this study is the introduction of:

- Redundant Binary (RB) representation, which retains carries
- Carryless arithmetic, which removes carries entirely

A binary integer $n$ is mapped to a polynomial:

$$\boldsymbol{n} = \sum_i b_i 2^i \qquad \leftrightarrow \qquad P(x) = \sum_i b_i x^i$$

The Shiftless update rule becomes, in the polynomial domain:

$$P_{k+1}(x) = (x+1)P_k(x) + x^{L_k}$$

where:

- $(x+1)P_k(x)$ represents geometric expansion (Reach)
- $x$^{$L_k$} represents LSB noise injection (Fill)

Evaluating the RB polynomial at $x = 2$ yields exactly the Shiftless integer value (**Figure 2**), proving that RB preserves all information. Projecting coefficients modulo 2 (i.e., onto GF(2)) removes carries and reveals the pure algebraic skeleton of the trajectory.

## 4. Emergence of the Sierpiński Gasket

Over GF(2), repeated multiplication by $(x+1)$ produces Pascal's triangle modulo 2, which is exactly Rule 90:

$$(x+1)^{2^m} = x^{2^m} + 1 \pmod 2$$

This identity guarantees perfect self-similarity: the initial configuration reappears shifted by $2^m$ bits.

Comparing standard $3^k$ (with carries) and carryless $(x+1)^k$ (without carries) reveals:

- Standard $3^k$: carries fill the gaps of the fractal and blur the structure
- Carryless: a perfect Sierpiński gasket emerges (**Figure 3**)

**Numerical Equivalence Verification (Seed: 27)**

| Step k | Standard Collatz | Shiftless (N) | N // LSB(N) | Status |
|---|---|---|---|---|
| 0 | 27 | 27 | 27 | MATCH |
| 1 | 41 | 82 | 41 | MATCH |
| 2 | 31 | 248 | 31 | MATCH |
| 3 | 47 | 752 | 47 | MATCH |
| 4 | 71 | 2272 | 71 | MATCH |
| 5 | 107 | 6848 | 107 | MATCH |
| 6 | 161 | 20608 | 161 | MATCH |
| 7 | 121 | 61952 | 121 | MATCH |
| 8 | 91 | 186368 | 91 | MATCH |
| 9 | 137 | 561152 | 137 | MATCH |
| 10 | 103 | 1687552 | 103 | MATCH |
| 11 | 155 | 5079040 | 155 | MATCH |
| 12 | 233 | 15269888 | 233 | MATCH |
| 13 | 175 | 45875200 | 175 | MATCH |
| 14 | 263 | 137887744 | 263 | MATCH |
| 15 | 395 | 414187520 | 395 | MATCH |
| 16 | 593 | 1243611136 | 593 | MATCH |
| 17 | 445 | 3732930560 | 445 | MATCH |
| 18 | 167 | 11207180288 | 167 | MATCH |
| 19 | 251 | 33688649728 | 251 | MATCH |
| 20 | 377 | 101200166912 | 377 | MATCH |
| 21 | 283 | 303868936192 | 283 | MATCH |
| 22 | 425 | 912680550400 | 425 | MATCH |
| 23 | 319 | 2740189134848 | 319 | MATCH |
| 24 | 479 | 8229157339136 | 479 | MATCH |
| 25 | 719 | 24704651886592 | 719 | MATCH |
| 26 | 1079 | 74148315398144 | 1079 | MATCH |
| 27 | 1619 | 222513665671168 | 1619 | MATCH |
| 28 | 2429 | 667678435966976 | 2429 | MATCH |
| 29 | 911 | 2003310185807872 | 911 | MATCH |
| 30 | 1367 | 6012129580679168 | 1367 | MATCH |
| 31 | 2051 | 18040786788548608 | 2051 | MATCH |
| 32 | 3077 | 54131156458668032 | 3077 | MATCH |
| 33 | 577 | 162411061562048512 | 577 | MATCH |
| 34 | 433 | 487514659662856192 | 433 | MATCH |
| 35 | 325 | 1463669878895411200 | 325 | MATCH |
| 36 | 61 | 4395513236313604096 | 61 | MATCH |
| 37 | 23 | 13258597302978740224 | 23 | MATCH |
| 38 | 35 | 40352252661239644160 | 35 | MATCH |
| 39 | 53 | 122209679488325779456 | 53 | MATCH |
| 40 | 5 | 368934881474191032320 | 5 | MATCH |
| 41 | 1 | 1180591620717411303424 | 1 | MATCH |

**Figure 1. Shiftless Model *vs* Standard Collatz (Odd Subsequence).**
*The Shiftless model preserves all bit information and matches the standard Collatz odd subsequence exactly when normalized by LSB.*

**Z[x] Polynomial Equivalence Verification (Seed: 27)**

| Step k | Shiftless N (Integer) | True RB P(2) | Max Coeff in P(x) | Status |
|---|---|---|---|---|
| 0 | 27 | 27 | 1 | MATCH |
| 1 | 82 | 82 | 2 | MATCH |
| 2 | 248 | 248 | 5 | MATCH |
| 3 | 752 | 752 | 8 | MATCH |
| 4 | 2272 | 2272 | 15 | MATCH |
| 5 | 6848 | 6848 | 29 | MATCH |
| 6 | 20608 | 20608 | 55 | MATCH |
| 7 | 61952 | 61952 | 108 | MATCH |
| 8 | 186368 | 186368 | 213 | MATCH |
| 9 | 561152 | 561152 | 414 | MATCH |
| 10 | 1687552 | 1687552 | 823 | MATCH |
| 11 | 5079040 | 5079040 | 1617 | MATCH |
| 12 | 15269888 | 15269888 | 3181 | MATCH |
| 13 | 45875200 | 45875200 | 6291 | MATCH |
| 14 | 137887744 | 137887744 | 12308 | MATCH |
| 15 | 414187520 | 414187520 | 24446 | MATCH |
| 16 | 1243611136 | 1243611136 | 47689 | MATCH |
| 17 | 3732930560 | 3732930560 | 94994 | MATCH |
| 18 | 11207180288 | 11207180288 | 185049 | MATCH |
| 19 | 33688649728 | 33688649728 | 369348 | MATCH |
| 20 | 101200166912 | 101200166912 | 719085 | MATCH |
| 21 | 303868936192 | 303868936192 | 1437294 | MATCH |
| 22 | 912680550400 | 912680550400 | 2800436 | MATCH |
| 23 | 2740189134848 | 2740189134848 | 5598567 | MATCH |
| 24 | 8229157339136 | 8229157339136 | 10927381 | MATCH |
| 25 | 24704651886592 | 24704651886592 | 21829550 | MATCH |
| 26 | 74148315398144 | 74148315398144 | 42673024 | MATCH |
| 27 | 222513665671168 | 222513665671168 | 85201144 | MATCH |
| 28 | 667678435966976 | 667678435966976 | 166781356 | MATCH |
| 29 | 2003310185807872 | 2003310185807872 | 332862501 | MATCH |
| 30 | 6012129580679168 | 6012129580679168 | 652372827 | MATCH |
| 31 | 18040786788548608 | 18040786788548608 | 1301619946 | MATCH |
| 32 | 54131156458668032 | 54131156458668032 | 2553812896 | MATCH |
| 33 | 162411061562048512 | 162411061562048512 | 5094274397 | MATCH |
| 34 | 487514659662856192 | 487514659662856192 | 10004926924 | MATCH |
| 35 | 1463669878895411200 | 1463669878895411200 | 19954355661 | MATCH |
| 36 | 4395513236313604096 | 4395513236313604096 | 39224310377 | MATCH |
| 37 | 13258597302978740224 | 13258597302978740224 | 78222038002 | MATCH |
| 38 | 40352252661239644160 | 40352252661239644160 | 153885420810 | MATCH |
| 39 | 122209679488325779456 | 122209679488325779456 | 306857169565 | MATCH |
| 40 | 368934881474191032320 | 368934881474191032320 | 604121936864 | MATCH |
| 41 | 1180591620717411303424 | 1180591620717411303424 | 1204591966784 | MATCH |

**Figure 2. RB Polynomial Evaluation vs Shiftless Integer.**
*Evaluating P_k(2) matches the Shiftless trajectory exactly, confirming that RB preserves all information.*

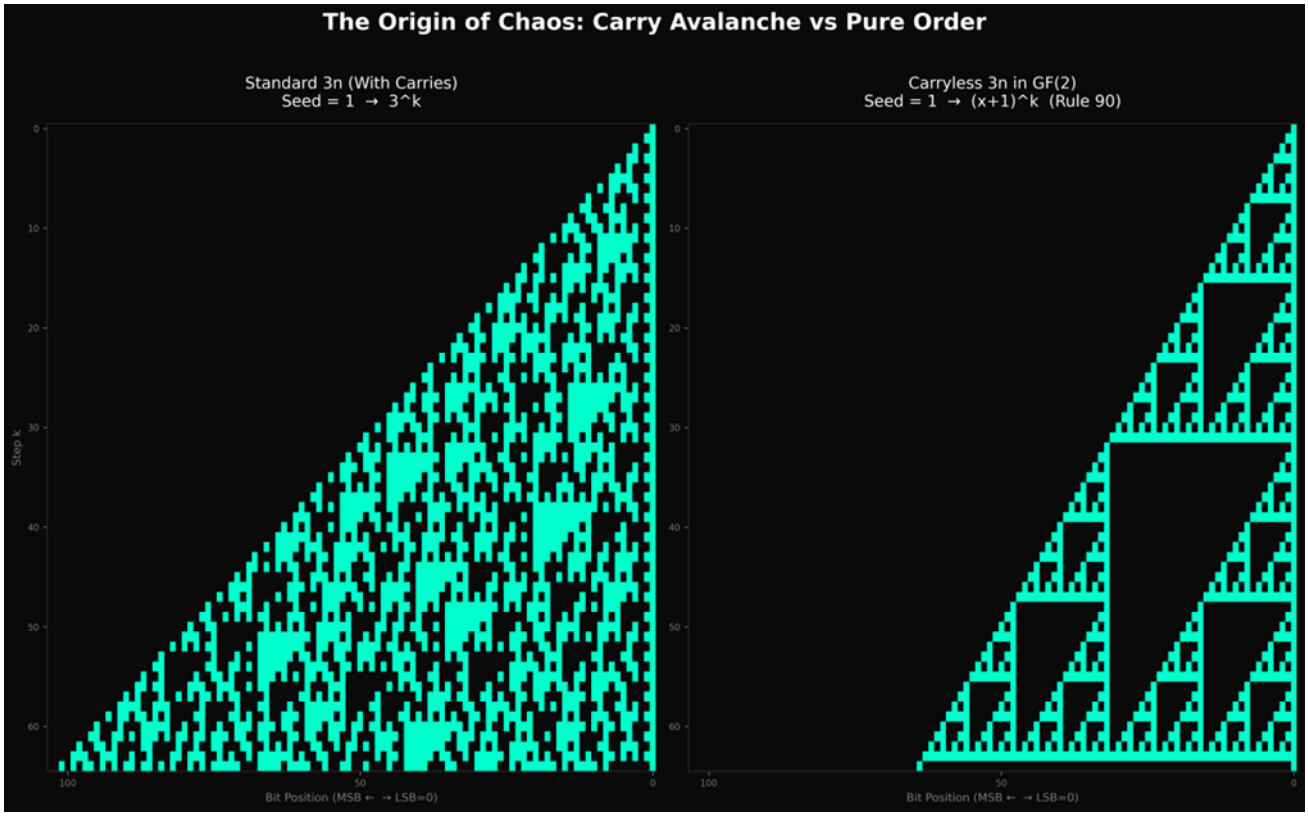


**Figure 3. Standard $3^n$ *vs* Carryless $(x+1)^n$.**
*Carry propagation destroys the fractal, while carryless arithmetic reveals the exact Sierpiński gasket.*

## 5. Emergent Chaos: Order × Noise × Carry

Using seeds 1 and 27, we compare:

- Shiftless model (noise + carry)
- Carryless Shiftless model (noise only)

The complexity of the Collatz trajectory arises from the interaction of:

1. Order:
   Fractal structure generated by repeated (x+1) (Rule 90)
2. Noise:
   LSB-dependent injection of $x^{L_k}$
3. Carry Avalanche:
   Noise enters the fractal gaps and triggers global structural distortion

Carryless Shiftless preserves the fractal skeleton, while Shiftless becomes chaotic due to carry avalanches (**Figures 4 and 5**).

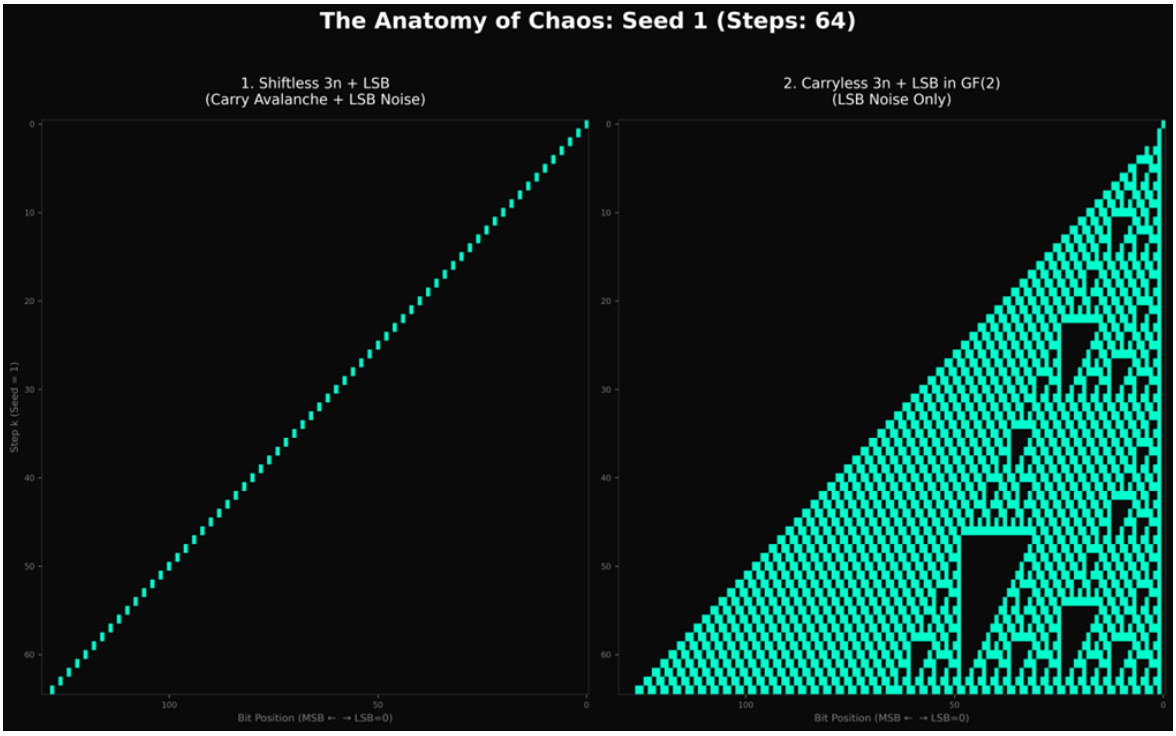


**Figure 4. Shiftless *vs* Carryless Shiftless (Seed = 1).**
*Noise-only dynamics preserve the fractal skeleton; carries destroy it.*

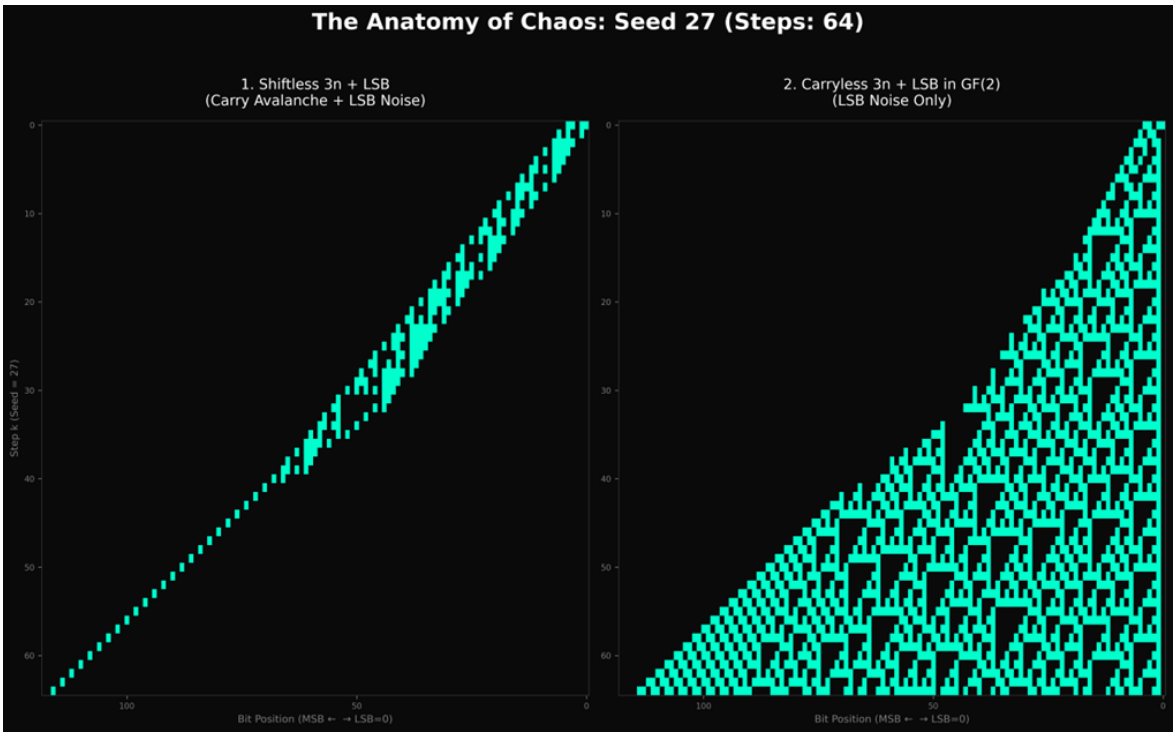


**Figure 5. Shiftless *vs* Carryless Shiftless (Seed = 27).**
*The same mechanism holds for nontrivial seeds: chaos emerges from the interference of order, noise, and carry.*

## 6. Conclusion

This study demonstrates that the Collatz trajectory is not merely a number-theoretic curiosity but a phenomenon at the intersection of:

- finite-field algebra
- fractal geometry
- information theory

The key insights are:

- The Shiftless model preserves all information and reveals the expansion structure.
- The RB polynomial representation records the full trajectory without losing carries.
- The Carryless model extracts the pure fractal skeleton (Rule 90).

Together, these models show that Collatz complexity emerges from the interference of fractal order, LSB noise, and carry avalanches. Future work includes analyzing RB coefficient distributions and the statistical behavior of LSB noise to gain deeper insight into long-term Collatz dynamics.

## License

MIT and CC BY 4.0
Copyright (c) Hiroshi Harada 2026